

Project Research Plan

Deep Learning Guided Discovery and FEP Validation of NOD2 Binders for the R702W Crohn's Variant - Tanmay Vasudeva

Rationale:

In the United States, inflammatory bowel disease (IBD) affects about 2.4 to 3.1 million people, and Crohn's disease affects about 1 million that can lead to a lifetime of chronic gastrointestinal irritation and significantly lower their quality of life. The NOD2 gene is currently the strongest genetic risk factor associated with Crohn's disease. The R702W mutation increases an individual's risk of developing Crohn's disease approximately 20 to 40 times more. The R702W mutation decreases the NOD2 protein's ability to recognize bacterial cell wall molecules which leads to an inappropriate response from the body's immune system and results in uncontrolled or excessive inflammation in the gut.

Despite evidence for the involvement of NOD2 in Crohn's disease, there are no FDA approved drugs precisely targeting this protein. The current treatment methods treat the subsequent inflammatory process as opposed to the causative agent and may eventually lose their effectiveness and result in severe side effects. Structural studies have shown that the leucine rich repeat (LRR) region of NOD2 has druggability potential but therapeutic development remains in its infancy. Since the safety profile of a drug is known when using drug repurposing as an approach to find new therapeutic uses for existing drugs it will be a more rapid method to focus on selecting candidates for potential future experimental trials. The in silico drug discovery strategy employed is aligned with the 3Rs (Replacement, Reduction and Refinement) since the primary use of high fidelity computer simulations to identify the best therapeutic candidates will eliminate the need for the use of vertebrate animals during this preliminary screening phase.

Research Question/Hypothesis(es)/Engineering Goal(s):

Research Question: Can existing compounds identified through a computational screen bind to the LRR domain of NOD2 and retain their binding affinities after the introduction of the R702W mutation associated with disease?

Hypothesis: Using a multi level computational strategy including molecular docking, machine learning, molecular dynamics and free energy perturbation (FEP) I plan to identify compounds that have been computationally determined to retain or increase their ability to bind to the NOD2 LRR domain within the R702W mutated state as opposed to the wild type form of NOD2; thus making them potential candidates for the development of compounds with reduced mutation sensitivity.

Expected Outcomes:

Selection of High Confidence Drug Leads: Selection of a number of lead drug candidates that have favorable NOD2 LRR binding properties based upon a high confidence assessment of their ability to bind NOD2 LRR from FDA approved and natural compounds library.

Quantification of Thermodynamics of Binding: Calculation of the effects of mutations on drug binding in a quantitative manner using a rigorous, gold standard method of Free Energy Perturbation (FEP) to calculate ΔG .

Selection of Mutation-Tolerant Molecules: Identification of drug candidates which are not affected by the mutation, thus allowing for treatment that are applicable with WT and is expected to demonstrate little variation in its binding with R702W.

Materials:

Computational Hardware:

Primary Workstation: AMD Ryzen processor, NVIDIA RTX 4060 Ti GPU (8 GB VRAM) for development and initial docking.

High-Performance Computing (HPC): VAST.ai GPU instances (NVIDIA 4090/H200) will be utilized for intensive Molecular Dynamics and FEP calculations. (These cloud GPUs are used only as commercially rented compute resources. There is no institutional lab access and no external mentorship or supervision through the computer provider. No proprietary or institution-provided datasets are used, only public datasets and student-generated results. This project is not conducted in a Regulated Research Institution (RRI) or as part of any industrial research program.)

Total Computational Load: Estimated at approximately 600 GPU-hours.

Software & Libraries:

Docking: GNINA 1.0 (Deep learning-based molecular docking with CNN scoring).

Simulation: OpenMM 8.2 (High-performance molecular dynamics).

Force Fields: OpenFF 2.1.0 Sage (Small molecule parameters) and AMBER ff14SB (Protein parameters).

Free Energy Analysis: pymbar 4.0 (Multistate Bennett Acceptance Ratio for free energy calculations).

Machine Learning: XGBoost (Gradient boosting framework for ligand re-ranking).

Cheminformatics/Visualization: RDKit (Molecule preparation), PyMOL (Structural visualization).

Data Sources:

AlphaFold Q9HC29 (<https://alphafold.ebi.ac.uk/entry/Q9HC29>); PDB 5IRM (<https://www.rcsb.org/structure/5IRM>);

e-Drug3D (<https://chemoinfo.ipmc.cnrs.fr/MOLDB/index.php>); COCONUT (<https://coconut.naturalproducts.net>)

Procedures:

Phase 1 - Structure Preparation & Grid Generation: First, I'll obtain the human NOD2 AlphaFold2 structure and validate with a model of the rabbit NOD2 X-ray structure (PDB ID: 5IRM) to verify the structural similarity of the two proteins. Next, I will prepare the protein for docking by adding hydrogen atoms and assigning the appropriate protonation states at pH 7.4. Finally, I will create a docking grid that is positioned in the center of the LRR domain binding pocket so that the specific mutation site is targeted during docking.

Phase 2 - Library Assembly & Curation: I will access approved drugs listed in e-Drug3D as well as a set of COCONUT natural product structures that are also fairly diverse, which I will then use to create 3-D conformer representations using RDKit and apply AM1-BCC partial charge models to all the molecular structures I have created. My library is expected to have approximately 9,000-10,000 compounds ready for screening.

Phase 3 - High-Throughput Virtual Screening: GPU-based docking for this project will be accomplished using GNINA. The output from GNINA will be used with CNN to score the output of the docking algorithm to select a smaller subset of potential lead compounds (i.e., top 5-10% based on CNN affinity). These will then be analyzed in detail to determine if they have sufficient binding affinity for use as a potential lead compound.

Phase 4 - Machine Learning Re-ranking (NOD2-Scout): I will train a scaffold-split cross-validation XGBoost Classifier model on 2048-Bit Morgan fingerprints using a scaffold-split cross-validation approach to minimize bias in the training data to identify a focused set of high confidence hits from the docked candidates that will likely exhibit stable binding.

Phase 5 - ADMET Filtering: I will use Lipinski's rule of five and predict the pharmacokinetic characteristics to ensure candidate drug-likeness and remove those that have poor ADME (absorption, distribution, metabolism and excretion) characteristics in order to simulate those with a favorable profile.

Phase 6 - Molecular Dynamics (MD) Simulations: I will create protein-ligand systems in OpenMM using TIP3P-FB water models at 0.15 M NaCl to mimic physiological conditions. I will then perform a 20 ns production molecular dynamics simulation on each of the two systems per candidate systems (wild type NOD2 and R702W mutant NOD2) using a 4fs time step with hydrogen mass repartitioning to evaluate the long-term structural stability of each complex

Phase 7 - Free Energy Perturbation (FEP): I will create a thermodynamic cycle that has 60 lambda windows for each of the compounds in order to determine the relative binding free energy of the mutations with respect to WT. In total, this system will be simulated for 1 ns per window using softcore potential functions, after which the simulation data will be analyzed via MBAR to determine $\Delta\Delta G = \Delta G_{R702W} - \Delta G_{WT}$, as well as to determine bootstrap uncertainties, in order to provide a measure of the magnitude of the binding affinity changes due to the mutation.

AI Tool Disclosure: Generative AI tools (e.g. Large Language Models) can potentially be used as a minor aid to help with debugging code and checking the syntax in the beginning phase of this project; all uses will be recorded in a logbook. The student researcher will perform all aspects of the experimental design, data collection, analysis, interpretation and writing of the results independently.

Risk Assessment:

This is a purely computational project involving no physical experiments, hazardous chemicals, biological agents, or vertebrate animals. All work will be performed on personal computers and cloud computing resources. Computational risks (data loss, software errors) will be mitigated through regular backups and version-controlled environments. All

data to be utilized in this study (including protein structures from UniProt/PDB and drug compounds from e-Drug3D) is publicly available, de-identified, and contains no personally identifiable information (PII). Therefore, this project does not meet the definition of human participant research. No specific safety measures are required.

Data Analysis:

I will rank the docking results using CNN affinity scores and display the positions of the complexes that have been docked in PyMol. To evaluate the performance of my machine learning models I will use ROC-AUC, Precision Recall, Confusion Matrices, and Scaffold-Split Cross Validation. I will measure the stability of the structures found through MD simulation through RMSD/RMSF Plots and Protein-Ligand Contact Occupancy Analysis. For Free Energy Calculations I will use MBAR to calculate the Gibbs Free Energy ΔG and its uncertainty. The difference in free energies ($\Delta\Delta G$) will allow me to quantify the effect of mutations on binding. I will test whether $\Delta\Delta G$ differs from 0 using a z-score, $z = \Delta\Delta G / SE(\Delta\Delta G)$, where $SE(\Delta\Delta G)$ is the MBAR or bootstrap standard error. For a two-sided test at $\alpha = 0.05$, I will treat $|z| > 1.96$ (approximately 2) as significant.

Bibliography:

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Additional Project Information: